

Unbiased Estimators & Confidence Intervals



Point estimates for the mean and variance

- 1 A sample of 8 measurements has values 12, 15, 11, 14, 13, 16, 12, 15. Find the unbiased point estimate of the population mean.
 - 2 Explain why the unbiased estimator of population variance divides the sum of squared deviations by $n - 1$ rather than n , using the formula $s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$.
-

Computing the unbiased sample variance

- 3 For the sample 4, 7, 9, 6, 9, find the unbiased estimate of the population variance.
-

Constructing a confidence interval for the mean

- 4 A sample of $n = 36$ has mean $\bar{x} = 52$ and population standard deviation $\sigma = 6$ (known). Construct a 95% confidence interval for the population mean, using $z^* = 1.96$.
 - 5 A sample of $n = 25$ has mean $\bar{x} = 80$ and unbiased sample standard deviation $s = 10$ (population standard deviation unknown, so a t -distribution is used). Construct a 90% confidence interval, using $t^* \approx 1.711$ (with $df = 24$).
-

Interpreting a confidence interval

- 6 A 95% confidence interval for a population mean is calculated as (48.2, 55.8). State the correct interpretation of this interval, and identify a common misinterpretation to avoid.
-

Sample size and interval width

- 7 A researcher wants to halve the margin of error of a confidence interval for the mean, keeping the confidence level and σ fixed. Explain how the required sample size n must change, using the formula for margin of error $E = z^* \frac{\sigma}{\sqrt{n}}$.
-

Determining the required sample size

- 8 A researcher wants a 95% confidence interval for a population mean with a margin of error no larger than 2, given a known population standard deviation $\sigma = 9$. Find the minimum required sample size, using $z^* = 1.96$ and rounding up to the nearest whole number.
-

Confidence intervals for a proportion

- 9 A survey of 200 randomly selected voters finds that 120 support a proposal. Construct a 95% confidence interval for the true population proportion, using $\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ with $z^* = 1.96$.
-

Comparing pooled vs. unbiased variance estimates

- 10 Two independent samples give unbiased variance estimates $s_1^2 = 16$ (from $n_1 = 10$) and $s_2^2 = 25$ (from $n_2 = 15$). Find the pooled variance estimate, using $s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$.
-

Effect of confidence level on interval width

- 11 Explain why a 99% confidence interval is always wider than a 95% confidence interval for the same sample data, referencing the critical value z^* used in each.
-

Constructing a confidence interval for a difference of means

- 12 Two independent samples give $\bar{x}_1 = 45$ (with $n_1 = 40$, $\sigma_1 = 8$) and $\bar{x}_2 = 40$ (with $n_2 = 50$, $\sigma_2 = 6$). Construct a 95% confidence interval for $\mu_1 - \mu_2$, using $z^* = 1.96$ and the standard error formula $\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$.
-

Assessing whether a claimed population mean is plausible

- 13 A machine is supposed to fill bottles with 500ml of liquid. A sample of $n = 30$ bottles has mean $\bar{x} = 497$ ml and sample standard deviation $s = 8$ ml. Construct a 95% confidence interval for the true mean fill volume (using $z^* \approx 1.96$ as an approximation for the t -distribution with $df = 29$), and use it to comment on whether the machine's target of 500ml is plausible.
- 14 Explain why increasing the sample size n , while keeping everything else fixed, makes a confidence interval narrower, referencing the standard error formula $\frac{\sigma}{\sqrt{n}}$.
-

Interpreting overlapping confidence intervals

- 15** Two independent 95% confidence intervals for population means are $(10, 14)$ and $(13, 17)$. Explain what can and cannot be concluded about whether the two population means are different, given that the intervals overlap.